

Jenkins-Practical

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1. CI/CD Pipeline Setup Using Jenkins, Maven, Docker and Pipeline Script

- Launch one Ubuntu EC2 instance to set up and configure Jenkins.

The screenshot displays the AWS Management Console interface for an EC2 instance. The top navigation bar shows the AWS logo, a search bar, and the user's profile (PoojaPrakash). The breadcrumb trail indicates the path: EC2 > Instances > i-0eee779f283178b9c. The main content area is titled 'Instance summary for i-0eee779f283178b9c (My Web Server)'. It features a 'Connect' button, an 'Instance state' dropdown menu showing 'Running', and an 'Actions' dropdown menu. The instance is updated less than a minute ago. The summary is organized into three columns. The first column contains: Instance ID (i-0eee779f283178b9c), IPv6 address (none), Hostname type (IP name: ip-10-0-7-4.eu-north-1.compute.internal), Answer private resource DNS name (none), Auto-assigned IP address (It is taking a bit longer than usual to fetch your data), and IAM role (none). The second column contains: Public IPv4 address (13.61.182.133), Instance state (Running), Private IP DNS name (ip-10-0-7-4.eu-north-1.compute.internal), Instance type (t3.micro), VPC ID (vpc-024f8931266570790), and Subnet ID (subnet-02c17cda1d166a85b). The third column contains: Private IPv4 addresses (10.0.7.4), Public DNS (ec2-13-61-182-133.eu-north-1.compute.amazonaws.com), Elastic IP addresses (It is taking a bit longer than usual to fetch your data), AWS Compute Optimizer finding (Opt-in to AWS Compute Optimizer for recommendations), and Auto Scaling Group name (none).

Instance ID i-0eee779f283178b9c	Public IPv4 address 13.61.182.133 open address	Private IPv4 addresses 10.0.7.4
IPv6 address -	Instance state Running	Public DNS ec2-13-61-182-133.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-10-0-7-4.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-10-0-7-4.eu-north-1.compute.internal	Elastic IP addresses It is taking a bit longer than usual to fetch your data
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address It is taking a bit longer than usual to fetch your data	VPC ID vpc-024f8931266570790 (MyVPC)	Auto Scaling Group name -
IAM role -	Subnet ID subnet-02c17cda1d166a85b (Public-Subnet)	

- Connect to the instance using SSH and install the necessary tools like java, maven, git and docker.
- Add the Jenkins user to the Docker group to allow Jenkins to run Docker commands

```

aws | Europe (Stockholm) | PoojaPrakash (7871-6932-0097) | PoojaPrakash
root@ip-10-0-7-4:/home/ubuntu# java -version
openjdk version "21.0.10" 2026-01-20
OpenJDK Runtime Environment (build 21.0.10+7-Ubuntu-124.04)
OpenJDK 64-Bit Server VM (build 21.0.10+7-Ubuntu-124.04, mixed mode, sharing)
root@ip-10-0-7-4:/home/ubuntu# mvn -version
Apache Maven 3.8.7
Maven home: /usr/share/maven
Java version: 21.0.10, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.14.0-1018-aws", arch: "amd64", family: "unix"
root@ip-10-0-7-4:/home/ubuntu# docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
root@ip-10-0-7-4:/home/ubuntu#

```

- This application will be built, tested, and containerized using Jenkins.

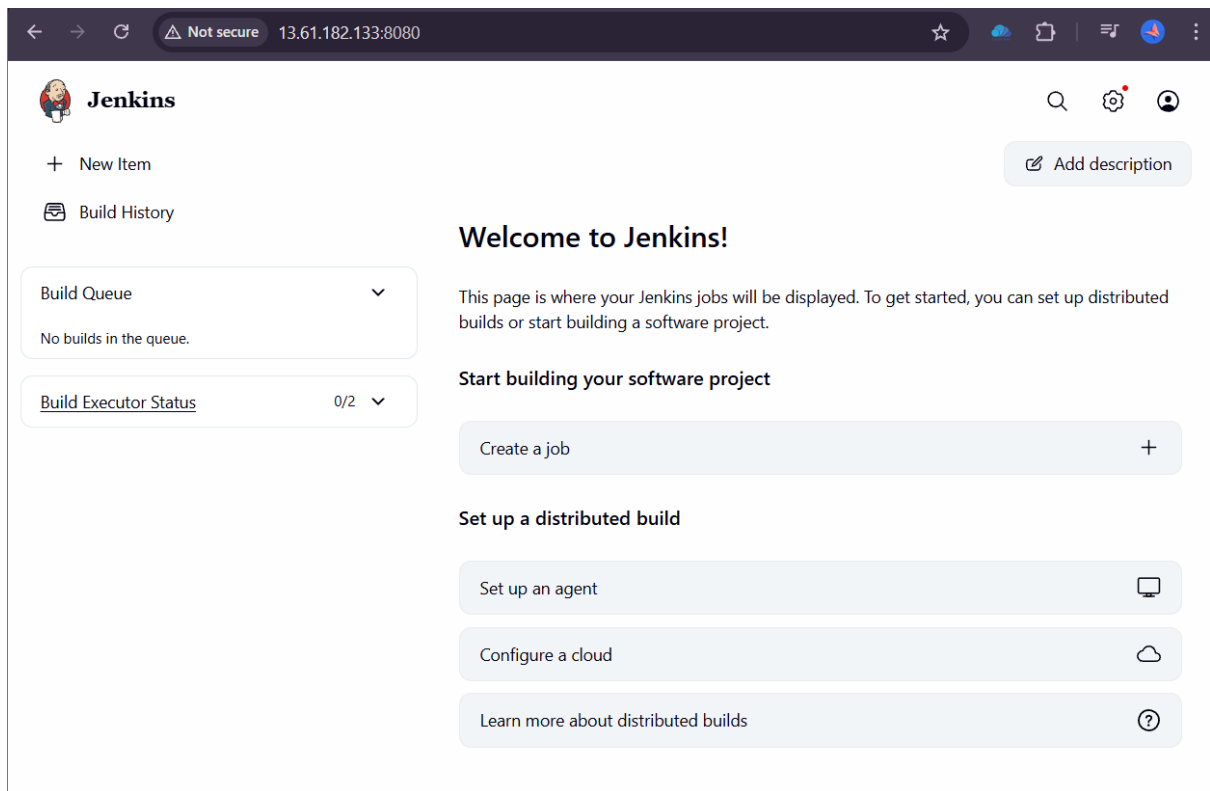
jenkins-tutorial / 3-java-jenkins-docker-app /

PoojaPrakash08 Remove unnecessary blank line in Dockerfile

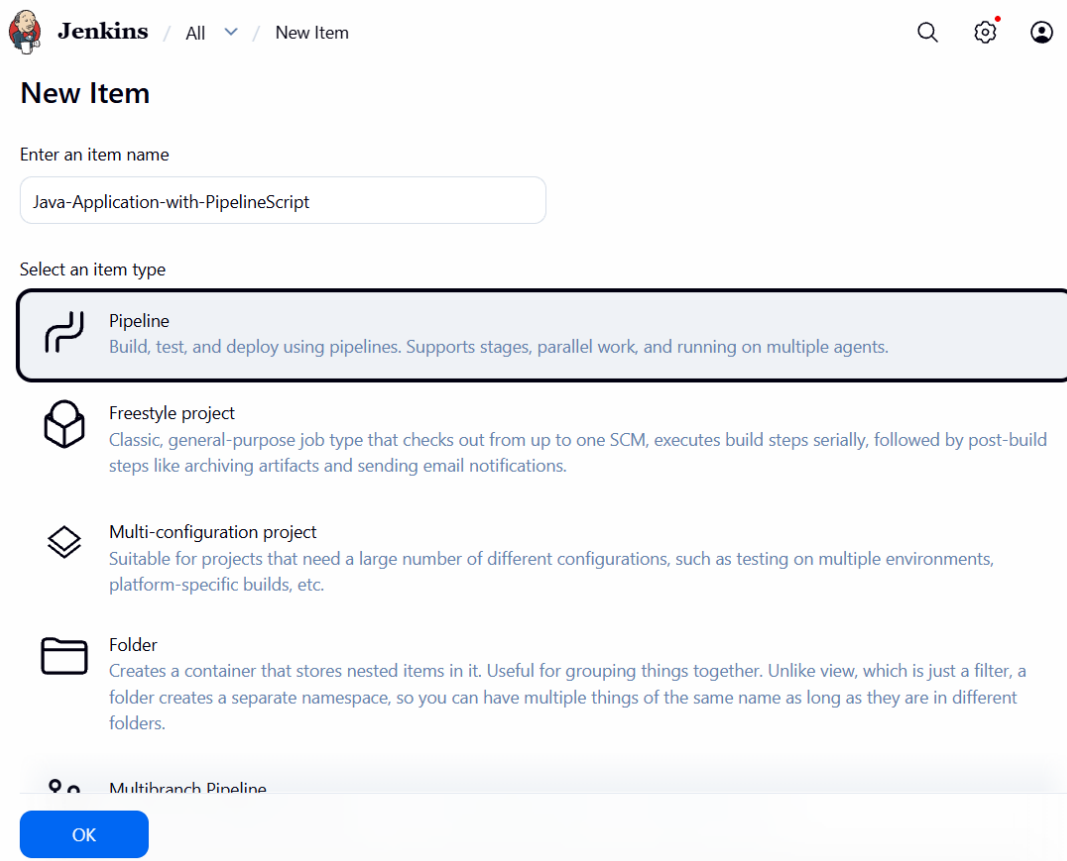
This branch is 1 commit ahead of theshubhamgour/jenkins-tutorial:main .

Name	Last commit message
..	
src	Direct-Struct mod
target	Direct-Struct mod
Dockerfile	Remove unnecessary blank line in Dockerfile
Jenkinsfile	Direct-Struct mod
README.md	Update README.md
docker-compose.yml	Direct-Struct mod
pipeline-jenkins-demo	Update pipeline-jenkins-demo
pom.xml	Direct-Struct mod

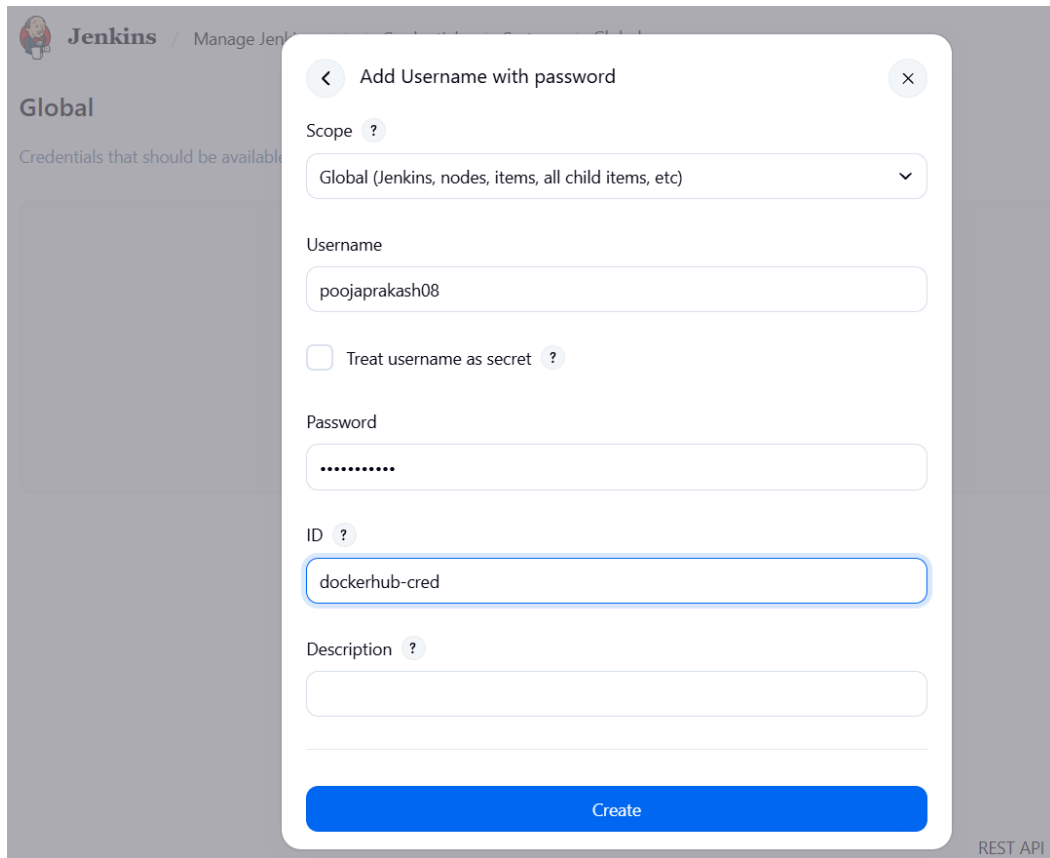
- Install Jenkins on the EC2 server and start the Jenkins service with 8080 port.



- Create a pipeline job with *java-App-with-PipelineScript* name



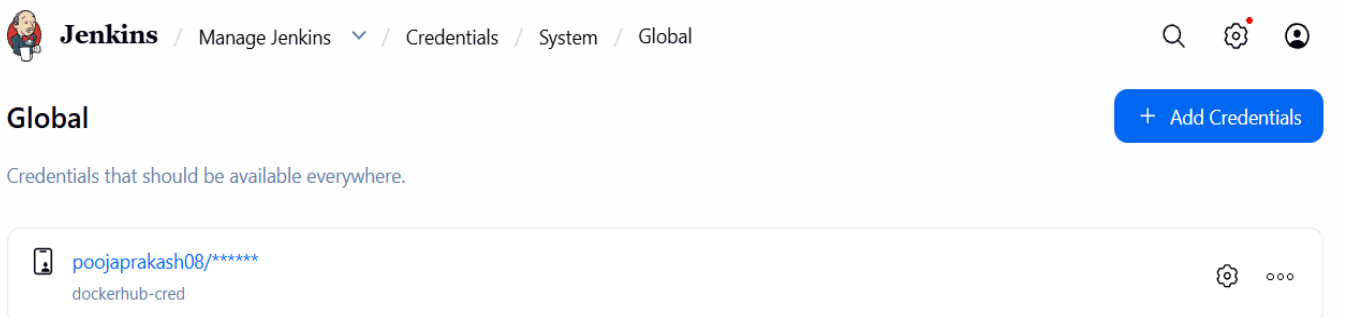
- Add the Docker Hub credentials in Global scope for secure authentication during Docker login and image push.



The screenshot shows the Jenkins 'Add Username with password' dialog box. The dialog is titled 'Add Username with password' and has a close button (X) in the top right corner. It contains the following fields and options:

- Scope**: A dropdown menu with the selected option 'Global (Jenkins, nodes, items, all child items, etc)'.
- Username**: A text input field containing 'poojaprakash08'.
- Treat username as secret**: A checkbox that is currently unchecked.
- Password**: A text input field with masked characters (dots).
- ID**: A text input field containing 'dockerhub-cred'.
- Description**: A text input field that is currently empty.

A blue 'Create' button is located at the bottom of the dialog. The background shows the Jenkins 'Global' credentials page with the text 'Credentials that should be available everywhere'.



The screenshot shows the Jenkins 'Global' credentials page. The page header includes the Jenkins logo and the breadcrumb 'Manage Jenkins / Credentials / System / Global'. On the right side, there are icons for search, settings, and a user profile. Below the header, the page title is 'Global' and the subtitle is 'Credentials that should be available everywhere.' A blue button labeled '+ Add Credentials' is located on the right. Below this, a list of credentials is shown, with the first entry being 'poojaprakash08/*****' with the ID 'dockerhub-cred'. To the right of this entry are icons for settings and a menu.

- Generating pipeline script syntax

 **Jenkins**

Java-App-with-PipelineScr...

▼

Pipeline Syntax

ⓘ Global Variables Reference

ⓘ Online Documentation

ⓘ Examples Reference

ⓘ IntelliJ IDEA GDSDL

Steps

Sample Step

git: Git

git ⓘ

Repository URL ⓘ

https://github.com/PoojaPrakash08/jenkins-tutorial.git

Branch ⓘ

main

Credentials ⓘ

- none -

☒ Include in polling? ⓘ

☒ Include in changelog? ⓘ

Generate Pipeline Script

git branch: 'main', url: 'https://github.com/PoojaPrakash08/jenkins-tutorial.git'

 **Jenkins**

Java-App-with-PipelineScr...

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Pipeline Syntax

ⓘ Online Documentation

ⓘ Examples Reference

ⓘ IntelliJ IDEA GDSDL

Steps

Sample Step

withCredentials: Bind credentials to variables

withCredentials ⓘ

Secret values are masked on a best-effort basis to prevent *accidental* disclosure. Multiline secrets, such as the contents of a SSH private key file, are not masked. See the inline help for details and usage guidelines.

Bindings

⋮ Username and password (separated) ⓘ

Username Variable ⓘ

DOCKER_USER

Password Variable ⓘ

DOCKER_PASS

Credentials ⓘ

poojaprakash08/*****

▼

+ Add

Credentials ?

poojaprakash08/*****

+ Add

+ Add

Generate Pipeline Script

```
withCredentials([usernamePassword(credentialsId: 'dockerhub-cred', passwordVariable:
'DOCKER_PASS', usernameVariable: 'DOCKER_USER')]) {
  // some block
}
```

- The Jenkinsfile clones the repository, builds the application using Maven, runs tests, logs in to Docker Hub using Jenkins credentials, builds the Docker image and runs the container.

```
jenkinsfile
1 pipeline {
2   agent any
3
4   stages {
5
6     stage('Git Clone') {
7       steps {
8         git branch: 'main', url: 'https://github.com/PoojaPrakash08/jenkins-tutorial.git'
9       }
10    }
11
12    stage('Build') {
13      steps {
14        dir('3-java-jenkins-docker-app') {
15          sh 'mvn clean package'
16        }
17      }
18    }
19
20    stage('Test') {
21      steps {
22        dir('3-java-jenkins-docker-app') {
23          sh 'mvn test'
24        }
25      }
26    }
27  }
```

```
main.tf  index.html  jenkinsfile  dockerfile  jenkinsfile1

jenkinsfile
1  pipeline {
4    stages {
28      stage('Docker Login') {
29        steps {
30          withCredentials([usernamePassword(
31            credentialsId: 'dockerhub-creds',
32            usernameVariable: 'DOCKER_USER',
33            passwordVariable: 'DOCKER_PASS'
34          )]) {
35            sh '''
36              echo $DOCKER_PASS | docker login -u $DOCKER_USER --password-stdin
37            '''
38          }
39        }
40      }
41
42      stage('Docker Build') {
43        steps {
44          dir('3-java-jenkins-docker-app') {
45            sh 'docker build -t java-jenkins-docker:latest .'
46          }
47        }
48      }
49
50      stage('Docker Run') {
51        steps {
52          dir('3-java-jenkins-docker-app') {
53            sh 'docker run --rm java-jenkins-docker:latest'
54          }
55        }
56      }
57    }
58  }
59
```



Jenkins

/ Java-App-with-PipelineScr...

/ Configure



Configure

General

Triggers

Pipeline

Advanced

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
1 pipeline {
2   agent any
3
4   stages {
5
6     stage('Git Clone') {
7       steps {
8         git branch: 'main', url: 'https://github.com/PoojaPr
9       }
10    }
11
12    stage('Build') {
13      steps {
14        dir('3-java-jenkins-docker-app') {
15          sh 'mvn clean package'
```

try sample Pipeline...

☒ Use Groovy Sandbox ?

Save

Apply



Jenkins

/ Java-App-with-PipelineScr...



Status

</> Changes

▶ Build Now

Configure

Delete Pipeline

Full Stage View

Stages

Rename

Pipeline Syntax

Java-App-with-PipelineScript

Adc

Stage View

Average stage times:
(full run time: ~22s)

#4
Feb 19
02:20
No
Changes

Git Clone	Build	Test	Docker Login	Docker Build	Docker Run
1s	8s	6s	1s	710ms	1s
1s	8s	6s	1s	710ms	1s

Permalinks

Builds >

...

Today

✓ #4 8:50 PM

▼